

A two-stage architecture for stock price forecasting by integrating self-organizing map and support vector regression

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Abstract(1/2)

Stock price prediction has attracted much attention from both practitioners and researchers. However, most studies in this area ignored the **non-stationary nature of stock price series**. That is, **stock price series do not exhibit identical statistical properties at each point of time**. As a result, the relationships between stock price series and their predictors are quite dynamic. It is challenging for any single artificial technique to effectively address this problematic characteristics in stock price series. **One potential solution is to hybridize different artificial techniques**. Towards this end, this study employs a two-stage architecture for better stock price prediction.

Abstract(2/2)

Specifically, the **self-organizing map(SOM)** is first used to decompose the whole input space into regions where data points with similar statistical distributions are grouped together, so as to contain and capture the non-stationary property of financial series. After decomposing heterogeneous data points into several homogeneous regions, **support vector regression(SVR)** is applied to forecast financial indices. The proposed technique is empirically tested using stock price series from seven major financial markets. The results show that the performance of stock price prediction can be significantly enhanced by using the two-stage architecture in comparison with a single SVR model.

Previous research result

Research	Comparison algorithms	Experimental data	Results
Tay and Cao (2001a)	Back-propagation neural network (BPN) and SVR	Futures and Bonds	SVR forecasts better than the BPN
Kim (2003)	BPN and SVR	Korea composite stock price index	SVR forecasts better than the BPN in terms of movement direction
Huang et al. (2005)	SVR, Linear Discriminant analysis, Quadratic discriminant analysis, and Elman BPN	Japan NIKKEI 225 Index	SVR forecasts better than other techniques in terms of movement direction
Pai and Lin (2005)	Hybrid model of ARIMA and SVM	Ten company stocks	The hybrid model forecasts better than SVR and ARIMA
Wittkemper and Steiner (1996)	Neural networks whose topologies were optimized by genetic algorithm	Sixty-seven German stocks	The new topology can yield good forecast results

Most comparison results show that prediction performance of **SVR is better than that of ANN.**

- **SVRs** implement the structural risk minimization principle.
(To minimize the misclassification error or deviation from correct solution of the training data)
- **ANNs** use the empirical risk minimization principle.
(To minimize the upper bound of generalization error)

The two-staged architecture

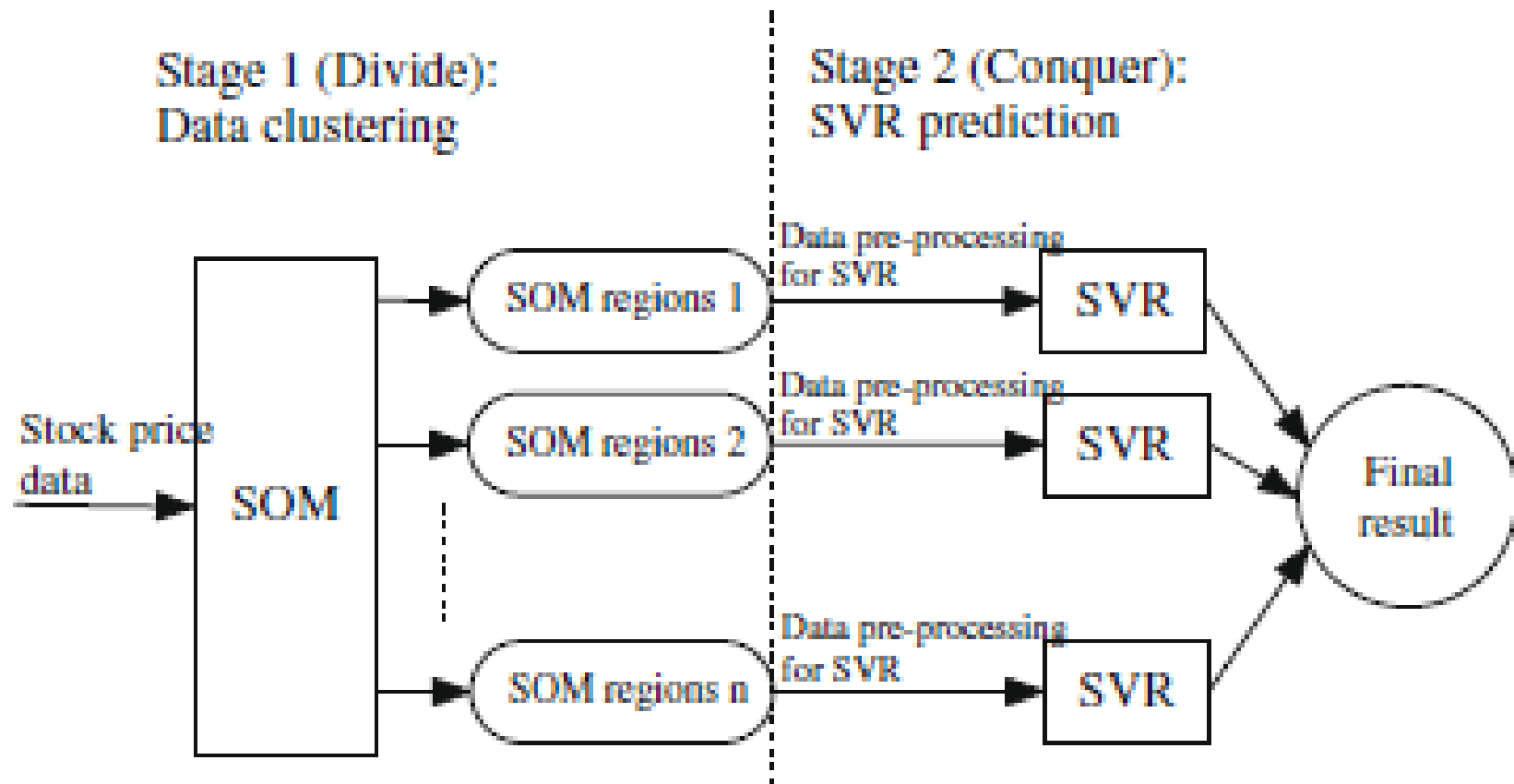


Fig. 4. The two-staged architecture.

Self Organizing Map(SOM)

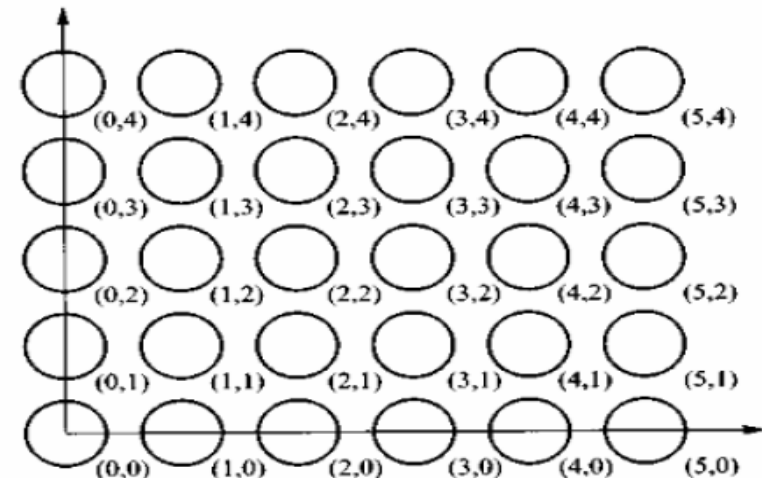
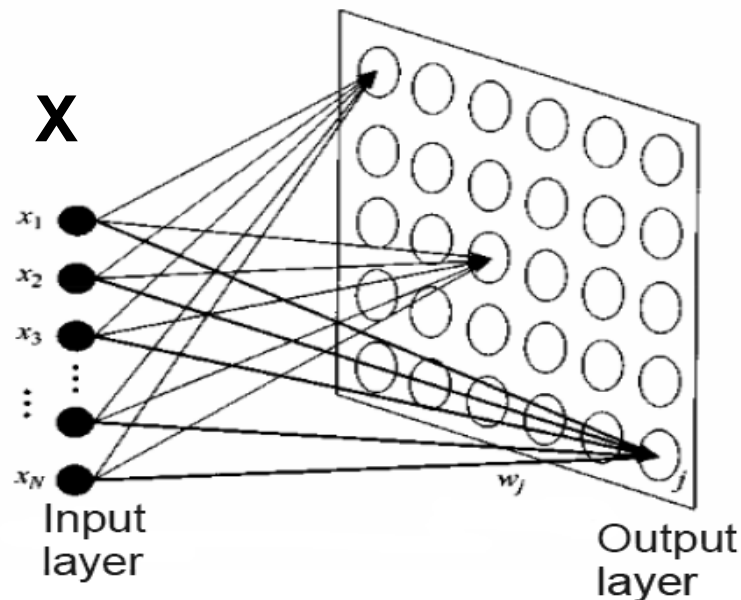
- SOM is an **unsupervised learning** method
- Clusters objects having multi-dimensional attributes into a lower-dimensional space.

- Goal:**

Maximize the degree of similarity of patterns within a cluster.

Minimize the similarity of patterns belonging to different clusters

- Input layer: $X = [x_1, x_2, \dots, x_N]^T$
- Output layer: $W_j = [w_{j1}, w_{j2}, \dots, w_{jN}]^T \quad j = \{(0,0), (1,0) \dots (5,4)\}$



- Center of the neighborhood

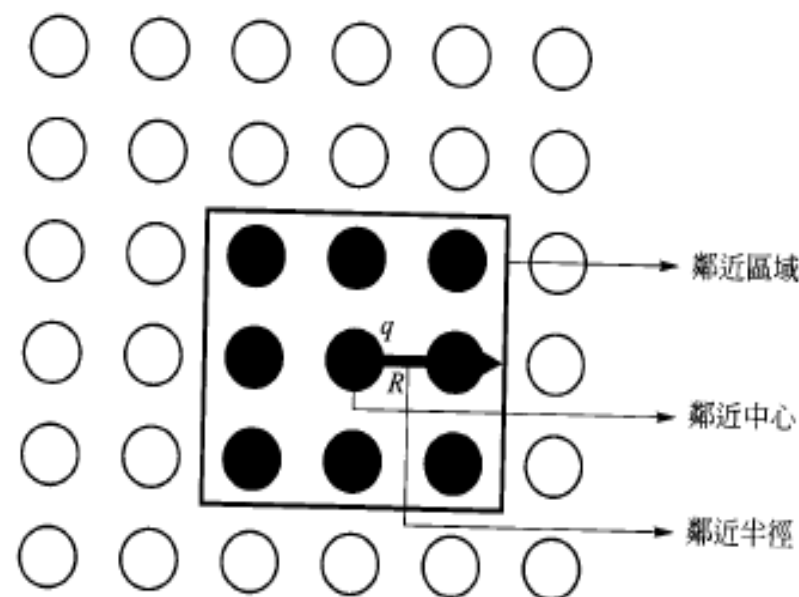
- winner node: $q(X) = \min_{\forall j} \|X - w_j\| = \sqrt{(X - w_{j1})^2 + \dots + (X - w_{jN})^2}$

- Neighborhood function

- $\eta_{qj} = \exp\left(\frac{-\|r_j - r_q\|^2}{R^2}\right)$ q: the winner R: neighbor radius

- Learning rule

- $w_j(k) = w_j(k-1) + \Delta w_j(k)$
 - $\Delta w_j(k) = \mu(k) \eta(k) [x(k) - w_j(k)]$
 - $\mu(k)$: the learning rate function at k



Begin

Set neighborhood parameters R

Set learning rate parameters η

Initialize weights randomly

While

For each input pattern x , compute the distance:

$$d_j \leftarrow \|x - w_j\|$$

Find index j such that d_j is a minimum (winning node)

For unit j and its neighborhoods, updates according to rules

$$\zeta(t) \leftarrow \eta(t) \exp(-d / R(t))$$

$$w_{jk} \leftarrow w_{jk} + \zeta(t) \times (x_k - w_{jk})$$

Reduce learning rate $\eta(t)$ and radius of $R(t)$ of neighborhood

Until (weight convergence or maximum no. of iterations is exceeded)

End

Fig. 3. SOM algorithm.

GHSOM implementation

- It can automatically determine the map size.
- It grows map size both in a hierarchical and horizontal way.

Parameter:

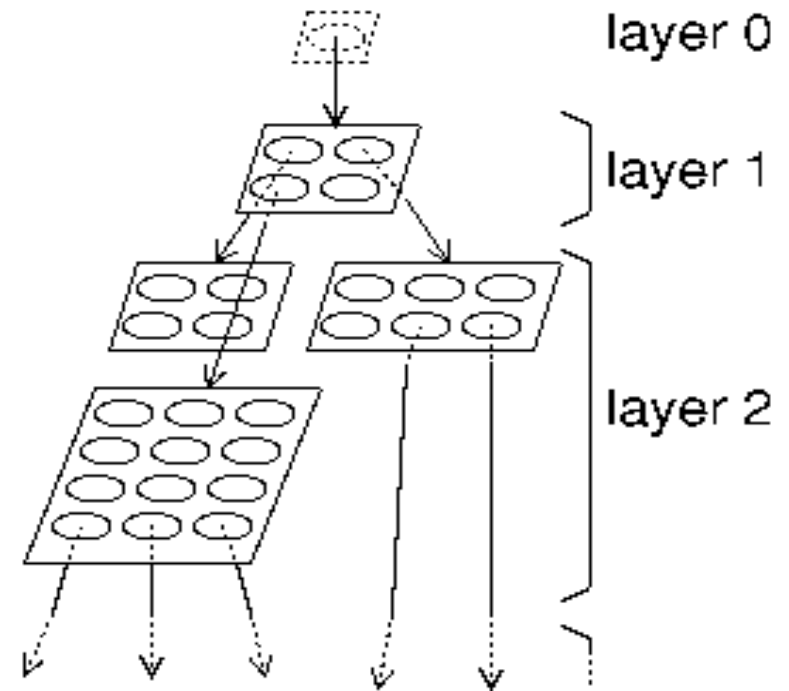
- Initial map size(layer 1) : 2x2
- Horizontal growing parameter: 0.05
- Hierarchical growing parameter: 1

Mean quantization error(unit):

$$mqe_i = \frac{1}{d} \|m_i - x\|$$

mqe_i : the mean quantization error of unit i

d : the number of input data contained in the unit i



Mean quantization error(map):

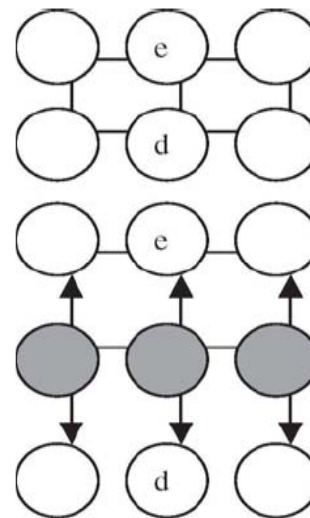
$$MQE_m = \frac{1}{u} \sum_i mqe_i$$

u : the number of units i contained in the SOM_m

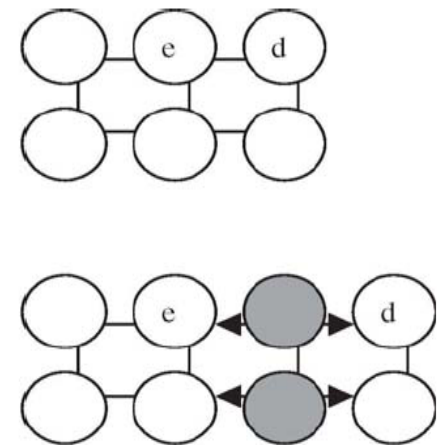
Horizontal expansion

after λ iterations if $MQE_m \geq \tau_1 MQE_0$, then expansion

- **Error unit:**
unit e with the highest MQE_e
- **Neighbor unit:**
unit d that is the most dissimilar to the error unit



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Hierarchical expansion

if $mqe_i \geq \tau_2 M\dot{Q}E_0$, then expansion

- The training process of the GHSOM is terminated when no more units require further expansion.

GHSOM Website:

<http://www.ifs.tuwien.ac.at/~andi/ghsom/>

Support Vector Regression(SVR)

SVR is closely related to SVM classifiers in terms of theory and implementation.

The output is a **real number**(SVM:output is a type of class)

$$f(x) = w^T x + b \text{ with } w \in X, \quad b \in R \quad (1)$$

$$\begin{aligned} \min \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & \begin{cases} y_i - w^T x_i - b \leq \varepsilon \\ w^T x_i + b - y_i \leq \varepsilon \end{cases} \end{aligned} \quad (2)$$

$$\begin{aligned} \min \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \\ \text{s.t.} \quad & \begin{cases} y_i - w^T x_i - b \leq \varepsilon + \xi_i \\ w^T x_i + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \end{aligned}$$

We set $\varepsilon = 0.001$ here

Implemented with libsvm:

<http://www.csie.ntu.edu.tw/~cjlin/libsvm/>

Data lie with this zone are deemed correct

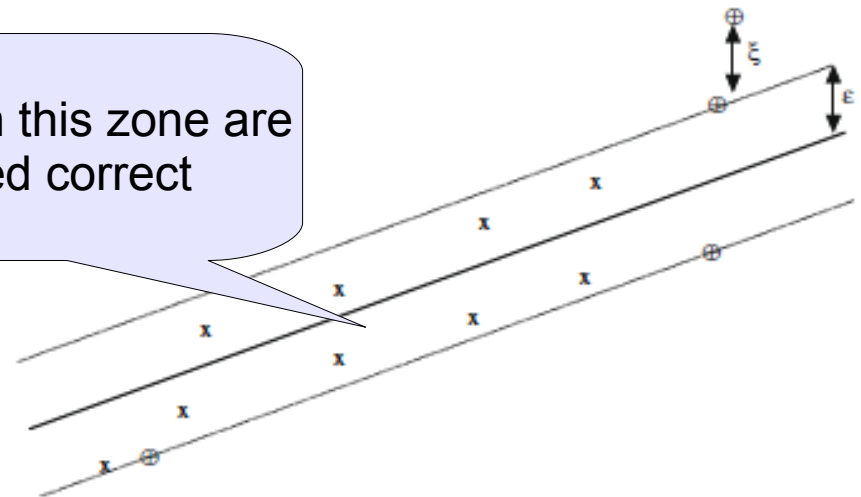


Fig. 1. Approximation function (solid line) of the SVR using an ε -insensitive zone (the area between dotted lines).

Experiment

- From July 1, 1997 to May 31, 2002
- seven stock market indices :
 1. Nikkei 225(NK, Japan, since 1950)
 2. All Ordinaries(AU, Australia, since 1980)
 3. Hang Seng(HS, Hong Kong, since 1969)
 4. Straits Times(ST, Singapore, since 1998)
 5. Taiwan Weighted(TW, Taiwan, since 1962)
 6. KOSPI(KO, Korea, since 1983)
 7. Dow Jones(DJ, USA, since 1896)

Daily closing prices were used

Closing price was transformed into four-lagged **relative difference in percentage of price(RDP)**, including **RDP-5**, **RDP-10**, **RDP 15**, and **RDP-20** and one transformed closing index(**EMA15**)

RDP : make the distribution of the data more symmetrical, thus **improving the predictive power** of artificial method.

Table 2

Input and output variables.

Input variables	Calculation
EMA15	$p(i) - \overline{EMA_{15}(i)}$
RDP-5	$(p(i) - p(i-5)) / p(i-5) * 100$
RDP-10	$(p(i) - p(i-10)) / p(i-10) * 100$
RDP-15	$(p(i) - p(i-15)) / p(i-15) * 100$
RDP-20	$(p(i) - p(i-20)) / p(i-20) * 100$
RDP+5	$\frac{\overline{p(i+5)} - \overline{p(i)}}{\overline{p(i)}} * 100$ $\overline{p(i)} = \overline{EMA_3(i)}$

- RDP values that lie beyond ± 2 standard deviations were first identified as outliers and the replaced with the close marginal values.
- 80% of the data was used for training, 20% for testing.
- Data were scaled into the range of $[-0.9, 0.9]$ to normalize each feature component

Performance criteria

Table 3

Performance metrics and their calculation.

Metrics		Calculation
NMSE	the smaller, the better	$\text{NMSE} = 1/(\delta^2 n) * \sum_{i=1}^n (a_i - p_i)^2$ $\delta^2 = 1/(n-1) * \sum_{i=1}^n (a_i - \bar{a})^2$
MAE	the smaller, the better	$\text{MAE} = 1/n * \sum_{i=1}^n a_i - p_i $
DS	the larger, the better	$\text{DS} = 100/n * \sum_{i=1}^n d_i$ $d_i = \begin{cases} 1 & (a_i - a_{i-1})(p_i - p_{i-1}) \geq 0 \\ 0 & \text{otherwise} \end{cases}$
WDS	the smaller, the better	$\text{WDS} = \sum_{i=1}^n d_i a_i - p_i / \sum_{i=1}^n d'_i a_i - p_i $ $d_i = \begin{cases} 0 & (a_i - a_{i-1})(p_i - p_{i-1}) \geq 0 \\ 1 & \text{otherwise} \end{cases}$ $d'_i = \begin{cases} 1 & (a_i - a_{i-1})(p_i - p_{i-1}) \geq 0 \\ 0 & \text{otherwise} \end{cases}$

NMSE, MAE : the deviation between actual and predicted values.

DS : the correctness of the predicted direction of RDP+5 in terms of percentage

WDS : the magnitude of the prediction errors as well as the direction

Result

Table 4

Prediction performance of the two-stage architecture and the single SVM.

		NMSE	MAE	DS	WDS
NK	SVR + SOM	1.165	0.216	55.80	0.847
	SVR	1.283	0.242	50.83	0.893
AU	SVR + SOM	1.078	0.132	53.36	0.850
	SVR	1.091	0.142	50.85	0.900
HS	SVR + SOM	0.913	0.101	59.07	0.805
	SVR	0.963	0.117	37.50	1.653
ST	SVR + SOM	0.949	0.082	53.94	1.002
	SVR	1.053	0.096	34.98	1.621
TW	SVR + SOM	1.006	0.156	51.35	0.957
	SVR	1.434	0.239	48.31	1.223
KO	SVR + SOM	1.013	0.143	53.53	1.095
	SVR	1.032	0.157	40.60	1.507
DJ	SVR + SOM	1.055	0.138	50.63	1.006
	SVR	1.186	0.160	47.35	1.088